

Flow Cytometry Analysis of Human Neutrophil Extracellular Traps (NETs) *in vitro*

Isolation of human neutrophils

1. Experiments using human neutrophils should be performed with the institutional approval and according to the guidelines of the Clinical Investigation Committee at the Facility.
2. Aspirate 10 to 15 mL of human peripheral blood from healthy volunteers by a 20 mL heparinized syringe, with their written or verbal consent.
3. Isolate human neutrophils using PolymorphaPrep™ density gradient media (#1114683; PROGEN Biotechnik GmbH, Heidelberg, Germany) according to the manufacturer's protocol.
4. Lay 15 mL aspirated blood over an equivalent volume of the density gradient medium in a 50 mL conical tube and centrifuge at 480 ×g and 20°C for 30 min.
5. Aspirate the upper layer of mononuclear cells, and transfer neutrophils in the lower layer to a 15 mL conical tube with 10 mL Ca²⁺, Mg²⁺ free Hank's balanced salt solution (#085-09355; FUJIFIM Wako Pure Chemical, Osaka, Japan).
6. Hemolyze with 2 mL diluted BD Pharm Lyse™ lysing buffer (10 x) (#55589; BD Biosciences, San Jose, CA, USA) with sterilized distilled water at room temperature (RT) for 5 min, after harvesting the cell pellet by centrifugation at 400 ×g and 20°C for 20 min.
7. Harvest neutrophils by centrifugation at 300 ×g and 4°C for 10 min, after making up the volume to 15 mL with phosphate buffer saline (PBS)- 2 mM ethylenediaminetetraacetic acid (EDTA) in the tube.
8. Suspend the isolated neutrophil pellet with 2 mL of 1% (v/v) fetal bovine serum (FBS)/RPMI-1640 (#187-02021; FUJIFIM Wako Pure Chemical) including penicillin–streptomycin antibiotics (#P4333; Sigma-Aldrich, St. Louis, MO, USA) in a 1:100 ratio.
9. Filter the suspended neutrophils through the 35 µm nylon mesh of a cell-strainer cap filter attached to a 5 mL polystyrene round bottom tube (# 352235; Corning, Oneonta, NY, USA).

10. Identify the suspended cells from the cellular fraction in the lower layer to comprise neutrophils on the basis of high positivity rates for neutrophil surface markers; CD16, CD11b, and CD66b etc., using flow cytometry.
11. Apply the cells in the lower layer as neutrophils for each *in vitro* assay.

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1. The assay protocol using flow cytometry is based on the literature; Masuda S, et al. Measurement of NET formation in vitro and in vivo by flow cytometry. *Cytometry A*. 2017; 91: 822-829. 10.1002/cyto.a.23169. PMID: 28715618.
2. Prepare each of the conditioned medium of 1% (v/v) FBS/RPMI-1640 with human Fgn (hFgn) (2 mg/mL) (#F4883; Sigma-Aldrich) and recombinant human tumor necrosis factor- α (rhTNF- α) (100 ng/mL) (#AF-300-01A, PeproTech, Rocky Hill, NJ) in the presence or absence of DF-521 (0.4 BU/mL) (Tobishi Pharmaceutical Co., Ltd., Tokyo, Japan).
3. Apply 500 μ L each of the conditioned media per well of a 24-well suspension plates (#662102; Greiner, Wagenseilgasse, Wien, Austria).
4. Adjust the isolated neutrophils in 1% (v/v) FBS/RPMI-1640 medium at 1×10^6 cells/500 μ L, and seed per well with 500 μ L of each conditioned medium.
5. Incubate the neutrophils in each conditioned medium in a 5% CO₂ incubator at 37°C for 4 h.
6. First, harvest each of the conditioned media containing non-adherent neutrophils into 15 mL conical tubes.
7. Next, harvest the adherent neutrophils by pipetting with 800 μ L PBS- 2 mM EDTA into the corresponding tubes, and assemble completely by repeating the procedure twice.
8. Centrifuge the harvested neutrophils at 220 $\times g$ and 4°C for 15 min.
9. Suspend the pellets in 200 μ L flow cytometry buffer (PBS- 2 mM EDTA containing 0.1% [v/v] FBS).
10. Filter the neutrophils suspended in each condition through a new similar 35 μ m nylon mesh of a cell-strainer cap filter put on a 1.5 mL capless graduated microcentrifuge tube (#NC-509-GRD-Q; Quality Scientific Plastics, San Diego, CA, USA).

11. Divide the cell suspension into two 100 μ L aliquots, after the addition of 2 μ L True Stain Monocyte Blocker™ (#426101; BioLegend, San Diego, CA, USA).
12. Stain the conditioned neutrophils in each 100 μ L aliquot of flow cytometry buffer with 2 μ L of mouse anti-human MPO antibody–FITC conjugate (#ab11729; clone 2C7; Abcam, Cambridge, UK) or an isotype control of BD Pharmingen™ Mouse IgG1 κ –FITC conjugate (#555909; clone MOPC-21; BD Biosciences) at 4°C for 30 min.
13. Wash the neutrophils in each aliquot twice with 800 μ L flow cytometry buffer at 800 $\times g$ and 4°C for 2 min.
14. Resuspend the stained neutrophils in 200 μ L flow cytometry buffer.
15. Add Sytox™ Orange (SO) (#S-11368; Molecular Probes, Eugene, OR, USA) at 5 μ M of final concentration after the dilution with flow cytometry buffer, and incubate at RT for 5 min, to stain the extracellular DNA of NETs.
16. Subject the stained neutrophils to flow cytometry in a BD FACSVerse™ flow cytometer (Becton Dickinson, Franklin Lakes, NJ, USA).
17. Analyze the data by FlowJo™ software (Ver10.4.1; Tomy Digital Biology CO., Ltd., Tokyo, Japan).
18. Repeat the assays with three different volunteers at least to obtain experimental replicates.